

Andrew Kerfoot

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PROFILE

Recent ecoinformatics graduate (GPA 3.94) with experience building reproducible Python and R workflows for ecological disease modeling, climate-data integration, and probabilistic forecasting. Applies scientific programming, quantitative analysis, and clear communication to data, research, software, and environmental analytics problems.

EDUCATION

B.S. Informatics - Ecology Emphasis | Northern Arizona University, Flagstaff, AZ

May 2026

GPA: 3.94 | Study Abroad: University of Liverpool, UK (Fall 2025)

TECHNICAL SKILLS

Programming: Python (Advanced) | R (Advanced) | SQL (Intermediate) | Java | C | HTML/CSS/JavaScript

Modeling & Analytics: Compartmental disease models | time-series forecasting (ETS, ARIMA, STL) | probabilistic forecasting | parameter sweeps | regression analysis

Data & Tools: Git/GitHub | Jupyter | RStudio | tidyverse/ggplot2 | matplotlib | Quarto/R Markdown | PRISM climate data | Microsoft Office | Google Workspace

Research: Reproducible workflows | geospatial and climate datasets | data visualization | scientific writing | experimental design

RESEARCH & PROJECT EXPERIENCE

Climate-Driven Disease Modeling - Undergraduate Principal Investigator | NAU

2025-2026

- Built a spatially explicit Python model in epymorph to simulate *Ambystoma tigrinum virus* (ATV) transmission across connected Arizona tiger salamander pond networks, testing how landscape connectivity shapes disease spread.
- Developed custom distance-based movement kernels and integrated daily PRISM climate data to represent seasonal adult migration and temperature-dependent transmission; scenario comparisons showed severe outbreaks emerge when susceptible larvae, carrier adults, and favorable temperatures overlap seasonally.
- Evaluated regional outbreak timing against historical ATV prevalence records and presented findings at OURCA Showcase, UGRADS, Undergraduate Symposium, and HURA culmination. [GitHub repository](#).

Influenza Forecasting - Time-Series Modeling Project

Spring 2026

- Built and compared six probabilistic influenza-like illness (ILI%) forecasting models in R using CDC FluSight quantile targets and hubverse-compatible outputs to support short-term influenza surveillance.
- Improved validation weighted interval score by 15% versus the untransformed STL+ETS baseline (0.299 vs. 0.350) while achieving 92% coverage for 90% prediction intervals; managed versioned outputs and submission through Git/GitHub.

ADDITIONAL EXPERIENCE

Field Research Technician - ATV Salamander Sampling | Northern Arizona University

Summer 2025

- Collected environmental swab and tissue samples from wild tiger salamander populations across Arizona pond sites, generating field data for ATV prevalence analysis and spatial disease-model development.
- Navigated remote field locations while maintaining sample integrity and chain-of-custody protocols.

CS 345 Grader | Northern Arizona University

Spring 2025

- Assessed student code submissions for correctness and quality, provided written feedback, and flagged academic-integrity concerns to faculty.